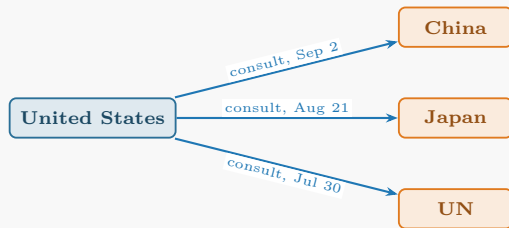


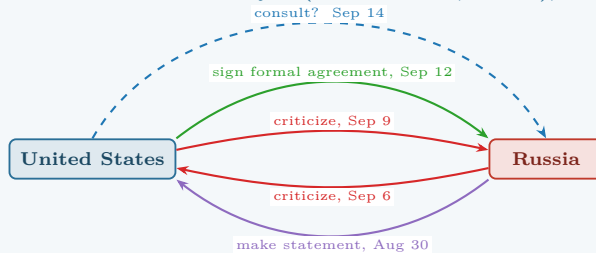
query: (United States, **consult**, ?, Sep 14) true answer: **Russia**

What a copy mechanism reads: (United States, consult, .)



candidates: China, Japan, UN. **Russia is not in this history**: the pair (United States, consult, Russia) never occurred, so every copy score is zero for it. On ICEWS18 this is the case for 23% of all test queries.

What ECHO reads: the dyad (United States, Russia), every relation, both directions



The dyad stream of (United States, Russia) is **agreement**, **criticize**, **criticize**⁻¹, **statement**⁻¹, with their times. A Transformer over this typed, ordered stream, read out for the query relation **consult**, gives Russia an intensity although no consult ever linked them.

ICEWS18 test queries by what the answer's history contains:

recurrent 50.6% (copy can see the answer)

dyad-only 23.4%

cold 26.0% (never interacted)